"Hashing"

$\mathbb{K}$ — Keyspace

$\frac{k}{m}$ = how full?

$\alpha = pk$ (from shuffeling)

load factor

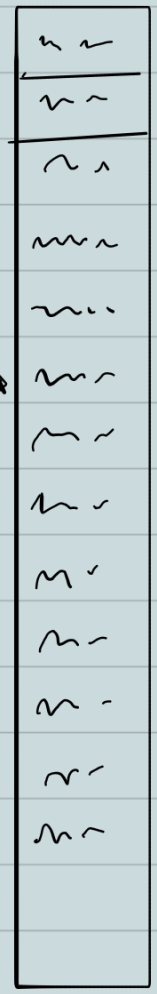
prob of k is in

$i \in [1..m] \equiv \{1, 2, ..., m\}$

$i \in \mathbb{K}$

Hash Table

d[] 1



m

Scatter function

hash function

hash

Access is O(1)

hashlike("test") → Add ascii values

→ int h

(if we start at 0)

h mod m +1

i

To fit into d[]

Subject to "Colisions" when hash(a) and hash(b) are the same

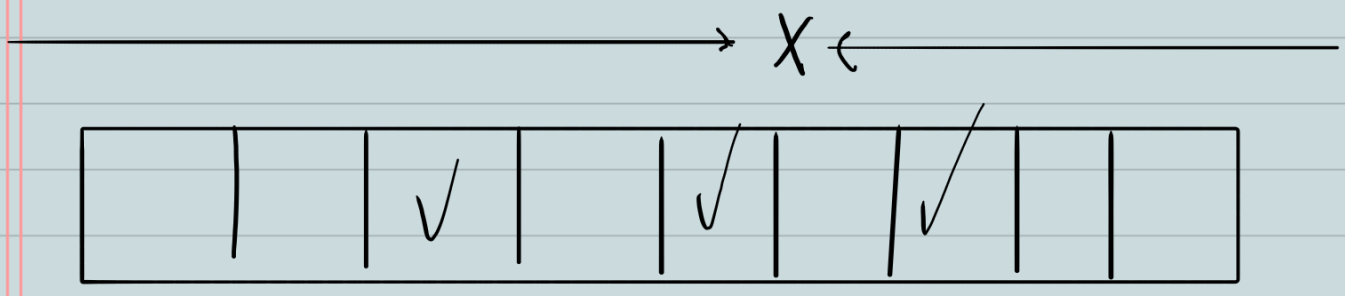
$i = (i + p) \bmod m + 1$

→ Prime no.

$$\alpha \le 80\% \Rightarrow \frac{1}{1-0.80} = 5$$

$$\alpha \le 90\% = \frac{1}{1-0.9} = 10$$ Search factor

$$\alpha \le 99\% = \frac{1}{1-0.99} = 100$$



$$\alpha = \frac{K}{m}$$

Loading factor

- 1 probe
- 2 probes $2\alpha(1-\alpha)$
- 3 probes $3\alpha^2(1-\alpha)$
- $k = K\alpha^{K-1}(1-\alpha)$

$$S_K = \sum K\alpha^{K-1}(1-\alpha)$$

$$= (1-\alpha)\sum K\alpha^{K-1}$$

$$T(n) = 1\alpha^0 + 2\alpha + 3\alpha^2 + \dots n\alpha^{n-1}$$

$$\alpha T(n) = \alpha + 2\alpha^2 + n\alpha^n$$

$$(1-\alpha)T(n) = 1 + \alpha + \alpha^2 + \alpha^3 \dots + \alpha^n + \alpha^{n-1}$$

$$U_n = 1 + \alpha + \alpha^2 + \dots \alpha^{n-1} - n\alpha^n$$

$$(1-\alpha)U_n = 1$$

$$= U_n = \frac{(1-\alpha^n)}{(1-\alpha)}$$

$$\lim_{n\to\infty} U_n = \frac{1}{1-\alpha}$$

$$(1-\alpha)T_N = \frac{\cancel{n\alpha^n}}{1-\alpha}$$

$$T_N = \frac{1}{(1-\alpha)^2}$$

$$S_n = (1-\alpha)T_N$$

$$S_N = \frac{1}{1-\alpha}$$

No n, so its independent of n. dependent on α.

$$\log_2(10\,90\,000) = 30$$

$$\frac{1}{1-0.5} = \frac{1}{0.5} = 5$$

Tutorial

Scatter Synchronization

Hashing slide

prac

linked lists

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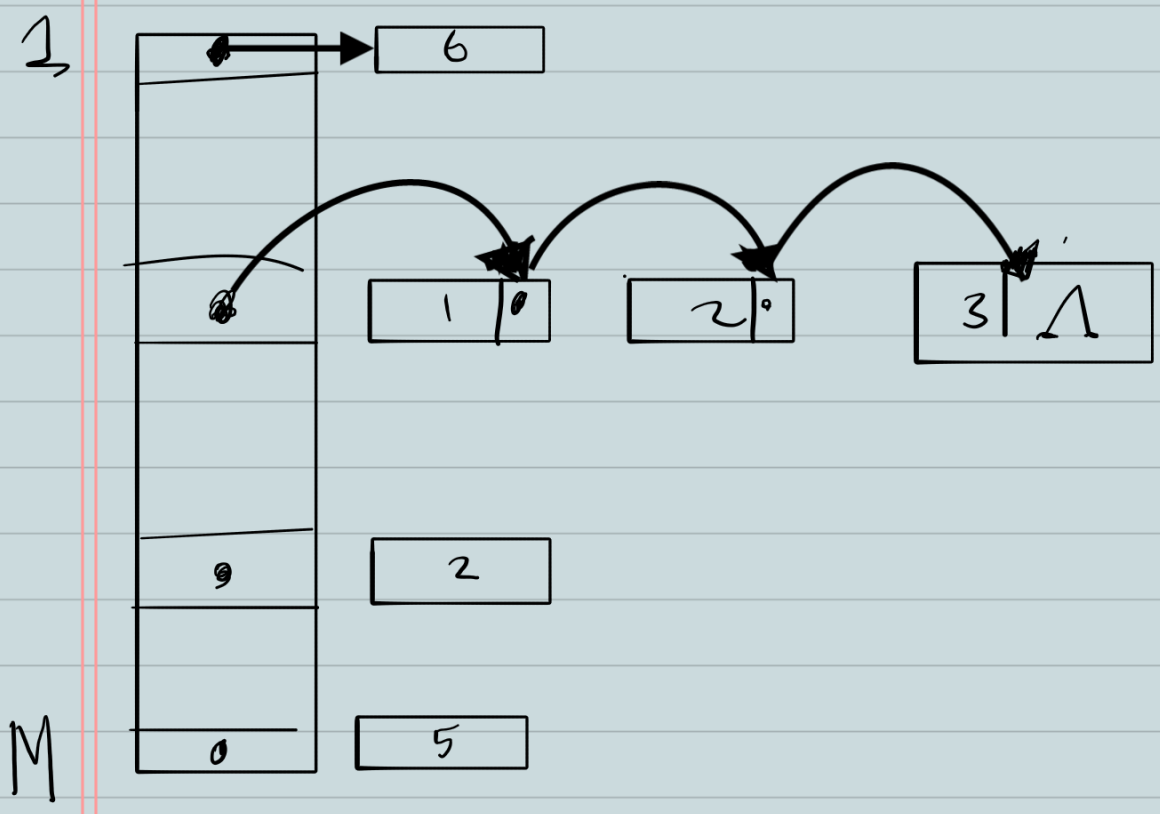
if → $\boxed{n > m}$

$i = h(key)$



prime: $6k \pm 1$

```
Node n = new Node();
```

Node

```
Node(int key, String value, Node pointer)
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